上海交通大學
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 6

Carbon Accounting: City Inventory

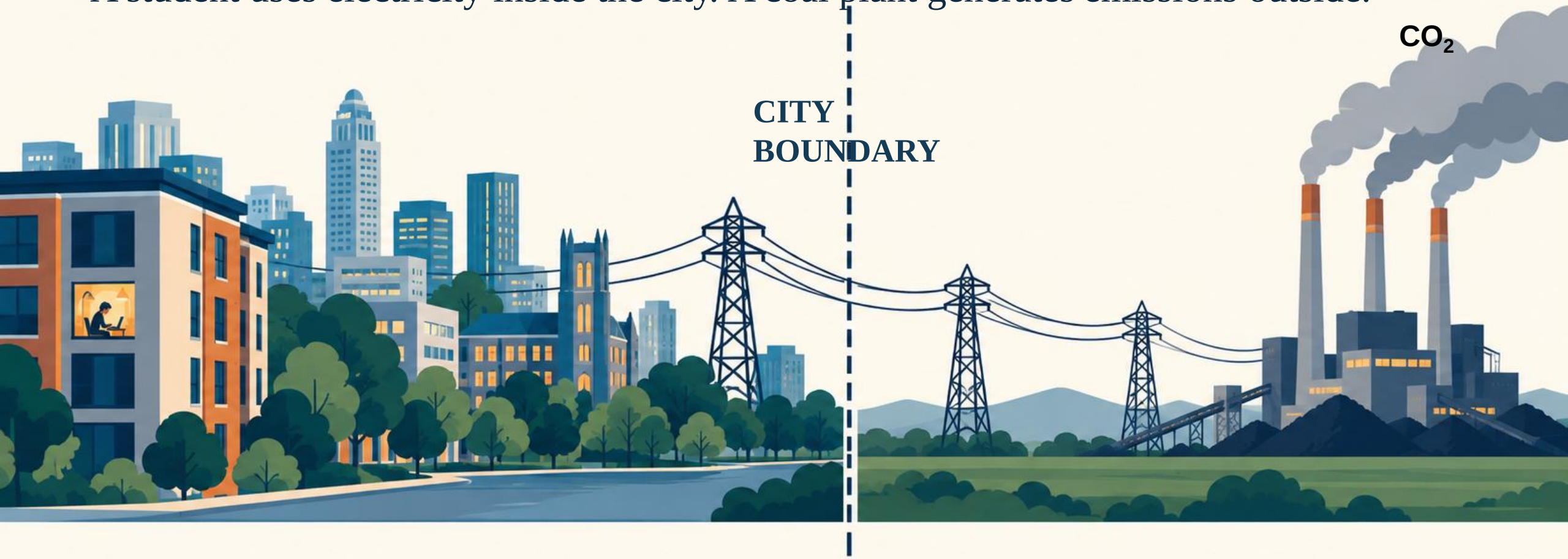
Haishan Yu

饮 水 思 源 · 爱 国 荣 校



Where Do a City's Emissions Occur?

A student uses electricity inside the city. A coal plant generates emissions outside.



Where would you place the emissions? Why?

投票 最多可选1项

设置

A student in City B uses electricity generated by a power plant in City A. Both cities prepare city-wide GHG inventories. How should the emissions be counted/reported?

A **City A only**, because the CO_2 is physically released there.

B **City B only**, because its electricity demand drives the generation.

C **Shared between the two cities**, based on their relative responsibility.

D **Reported by both cities**, but under different accounting categories.

E **Neither city**, because the emissions should only appear in the national inventory.

提交

Learning Outcomes

1. **Explain** why cities compile GHG inventories.
2. **Define** the boundary of a city inventory and **identify** cross-boundary activities that complicate emission attribution.
3. **Apply** the GPC framework to classify common urban emission sources by sector, scope and reporting level.

目 录

Contents

01- Why City GHG Inventories

Urban vulnerability, capacity and motivations for carbon accounting

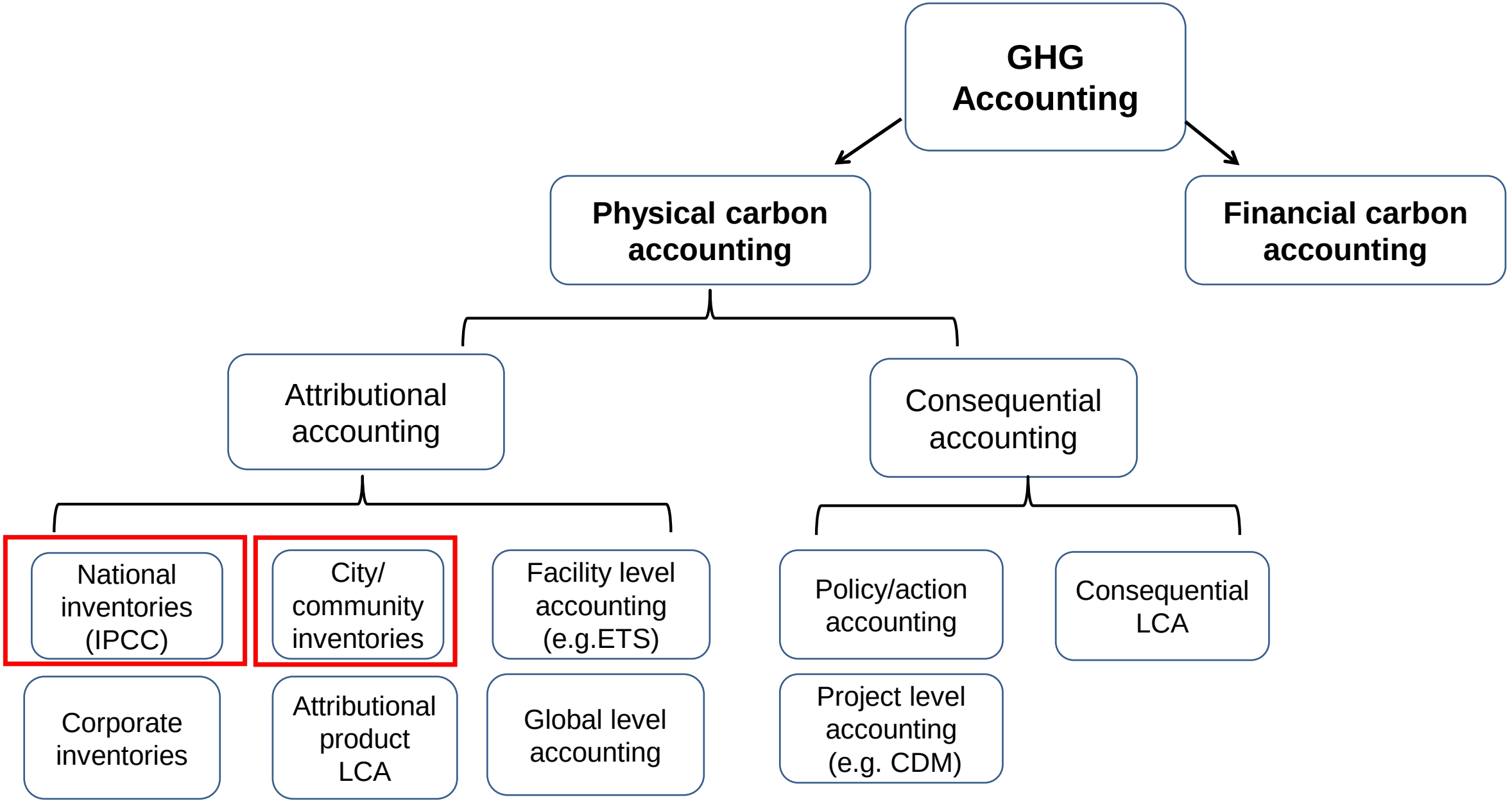
02- What Is Being Counted

Defining the city and tracing cross-boundary activities

03- How Are City Inventories Built

The GPC framework for sectors, scopes and reporting

04- In-class Quiz, Wrap up and Next time


GHG Accounting
Physical carbon accounting
Financial carbon accounting
Attributional accounting
Consequential accounting
National inventories (IPCC)
City/ community inventories
Facility level accounting (e.g.ETS)
Policy/action accounting
Consequential LCA
Corporate inventories
Attributional product LCA
Global level accounting
Project level accounting (e.g. CDM)

01

Why City GHG Inventories

Urban vulnerability, capacity and motivations for carbon accounting

Cities and climate change

- *"The effects of urbanization and climate change are converging in dangerous ways that seriously threaten the world's environmental, economic and social stability. "*

Global Report on Human Settlements 2011: Cities and Climate Change

Why Cities Matter in Climate Change

- Cities are:
  - **Drivers**
    - concentrated energy use, transport, buildings, industry, consumption
  - **Vulnerable systems**
    - heat, flooding, infrastructure risk, public health risk
  - **Solution spaces**
    - planning, public transport, buildings, waste, local energy systems


Cities are driving climate change. Here's how they can fix it
COUNTDOWN
TED

Cities are a big contributor

- Cities account for over 70 perc
  according to C40 Cities.
  - Population growth
  - A ravenous appetite for energy
- The world will continue to urba om
  56% in 2021 to 68% in 2050. (“ g
  the Future of Cities”)



Need help?

1/2

of the world's population live in cities.

80%

of global GDP is generated in cities.

2/3

of the world's energy is consumed by cities.

The global context | The risks | The opportunities

Need help?

$140bn

total annual damage on average between 2005 and 2014 as a result of climate change.

"Climate change poses a serious threat to the wealth and wellbeing of our city. Delaying action will be costly, which is why we are taking steps now to ensure we build resilience in Kisumu. CDP is the best initiative on climate change I have seen in a long time because it is focused on practical actions." - Kisumu,

$158tn

worth of assets at risk from climate change by 2050. That's double the total annual output of the global economy.

Cities are vulnerable to climate change

- 70% of cities are already dealing with the effects of climate change, and nearly all are at risk, according to C40 Cities.



Source: https://time.com/5687470/cities-countries-most-affected-by-climate-change/



Kiribati 基里



Haiti 海地



Yemen 也门



Manila 马尼拉



United Arab Emirate

Deaths

Power outages

Economic Impact

Financial damage

Infrastructure failures

Food and resource shortage

... ...



How Shanghai would look with a rise of just 2C: the UN warned this week of a potential 3C scenario. Photograph: Nickolay Lamm/Courtesy Climate Central

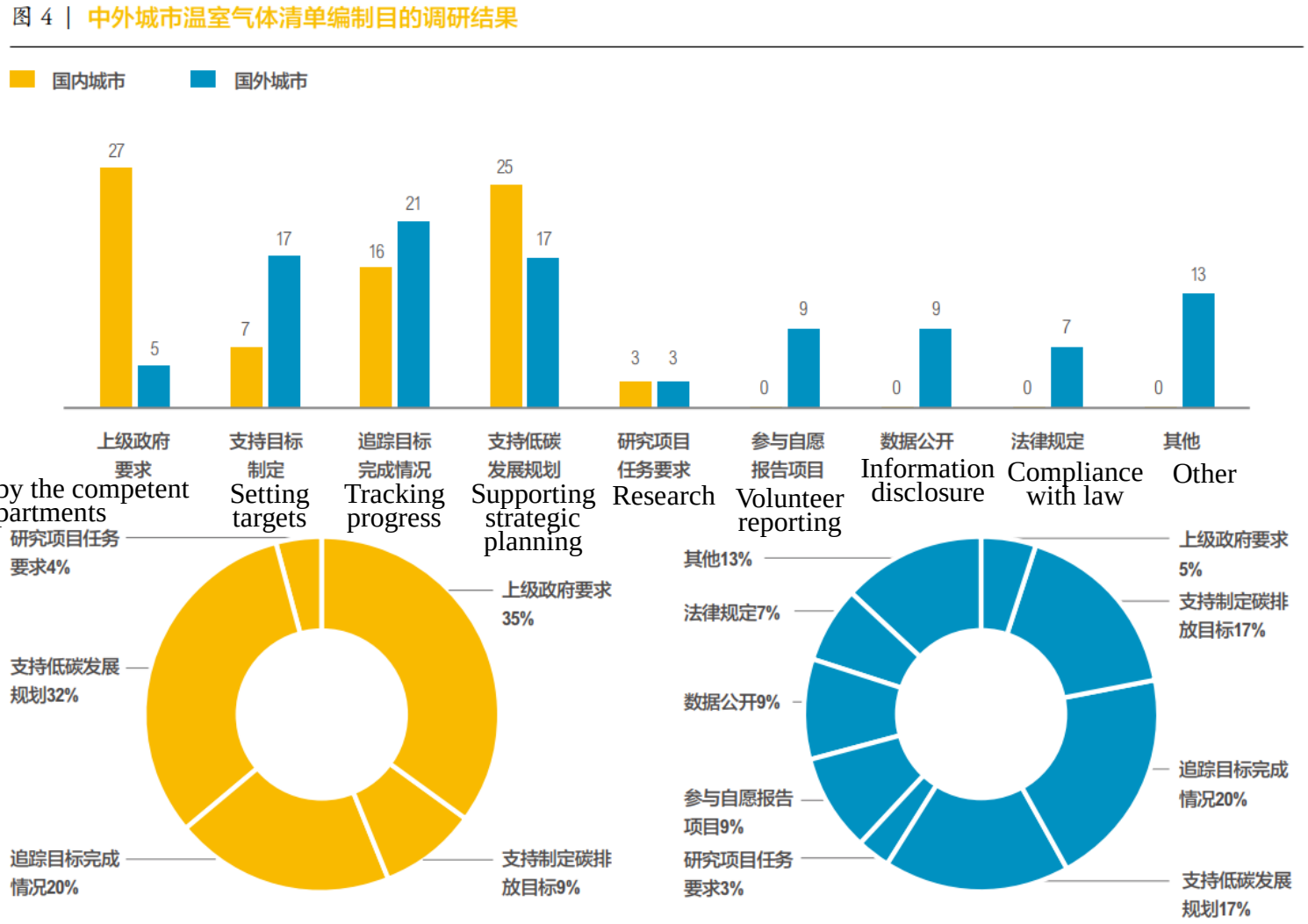
From: https://www.theguardian.com/cities/2017/nov/03/miami-shanghai-3c-warming-cities-underwater

Cities also have the opportunity to be integral to its solution

- Cities have the power to change the world.
- Urban density presents an opportunity to green life.
- Better in policy making (tailoring) and implementation compared to higher level of governance.
- …

Discussion

Why do cities need GHG inventories?



Source: 中国城市温室气体清单应用的领域、挑战和建议. 北京：世界资源研究所.
https://wri.org.cn/research/uses-challenges-and-recommendations-application-city-greenhouse-gas-inventories-china

Pre-class Exercise Summary

- For the city you checked before class, what did you find?

  A. A clear city-wide GHG inventory or inventory report
  B. A carbon-peaking or low-carbon plan with some emissions information, but no clear inventory
  C. Only scattered energy or emissions data
  D. No reliable public information found

  6B, 2A, and 2C out of 10 answers.

If a city has a climate target but no regular GHG inventory, what problems may arise?



从评估结果来看，试点城市的平均得分为79.2分，最高得分为93.7分，最低得分为68.9分。分领域得分方面，“基础工作与能力建设”和“低碳发展基本进展”平均得分率较高，分别为86.8%和86.1%，表明低碳试点城市在低碳发展基础能力建设方面开展了扎实工作并取得积极进展，并基本能够在实现预期经济增长目标的同时，实现碳排放总量的有效控制，并较好完成碳排放强度下降目标任务。而“创新举措”和“低碳发展体制机制建设”平均得分率略低，分别为72.1%和68.4%，表明多数试点城市的低碳发展创新力度不够，并普遍在低碳发展体制机制建设方面存在提升空间。

评估结果为优良的城市：北京、深圳、烟台、潍坊、衢州、常州、重庆、上海、济南、赣州、广州、合肥、安康、镇江、成都、杭州、济源、嘉兴、吉安、淮安、南昌、金华、三明、厦门、遵义、大兴安岭、青岛、南京、长阳土家族自治县、郴州、敦煌、昆明、贵阳、株洲、天津、吴忠、黄山、湘潭、温州、秦皇岛。

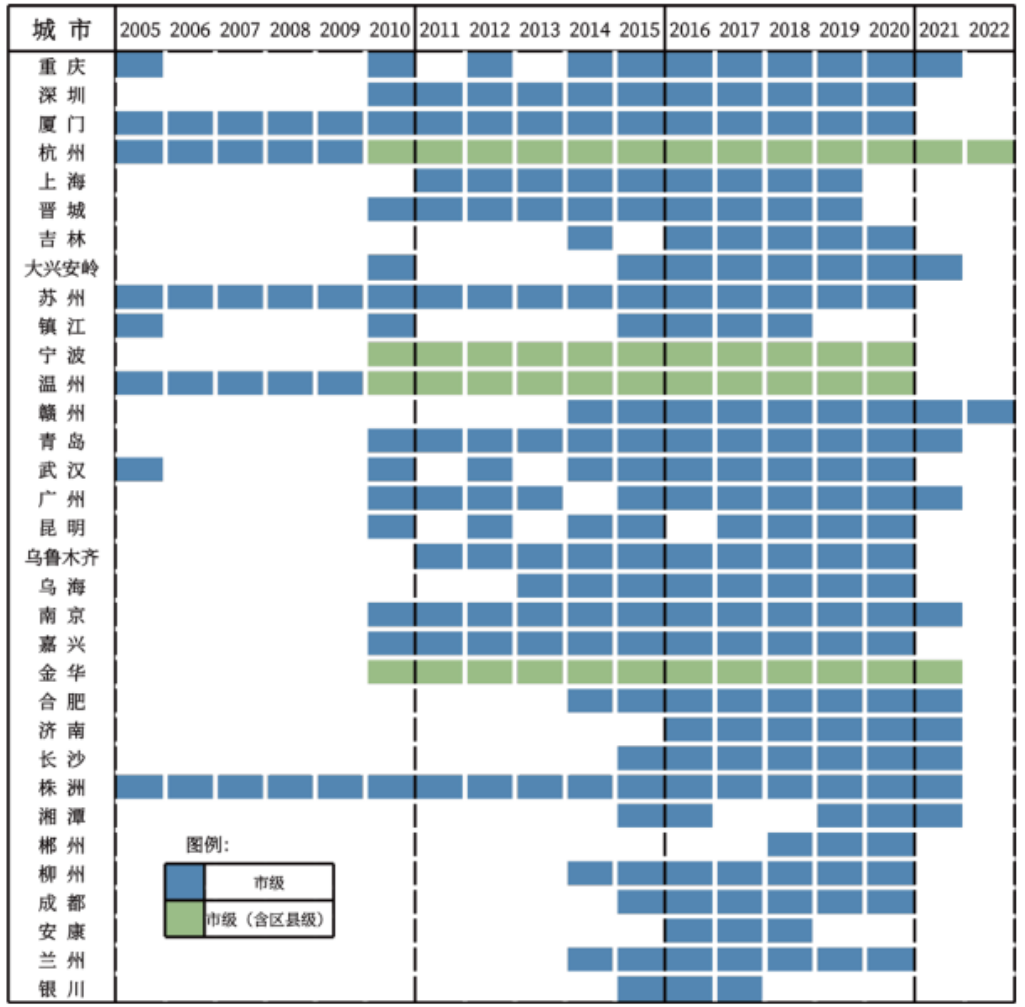
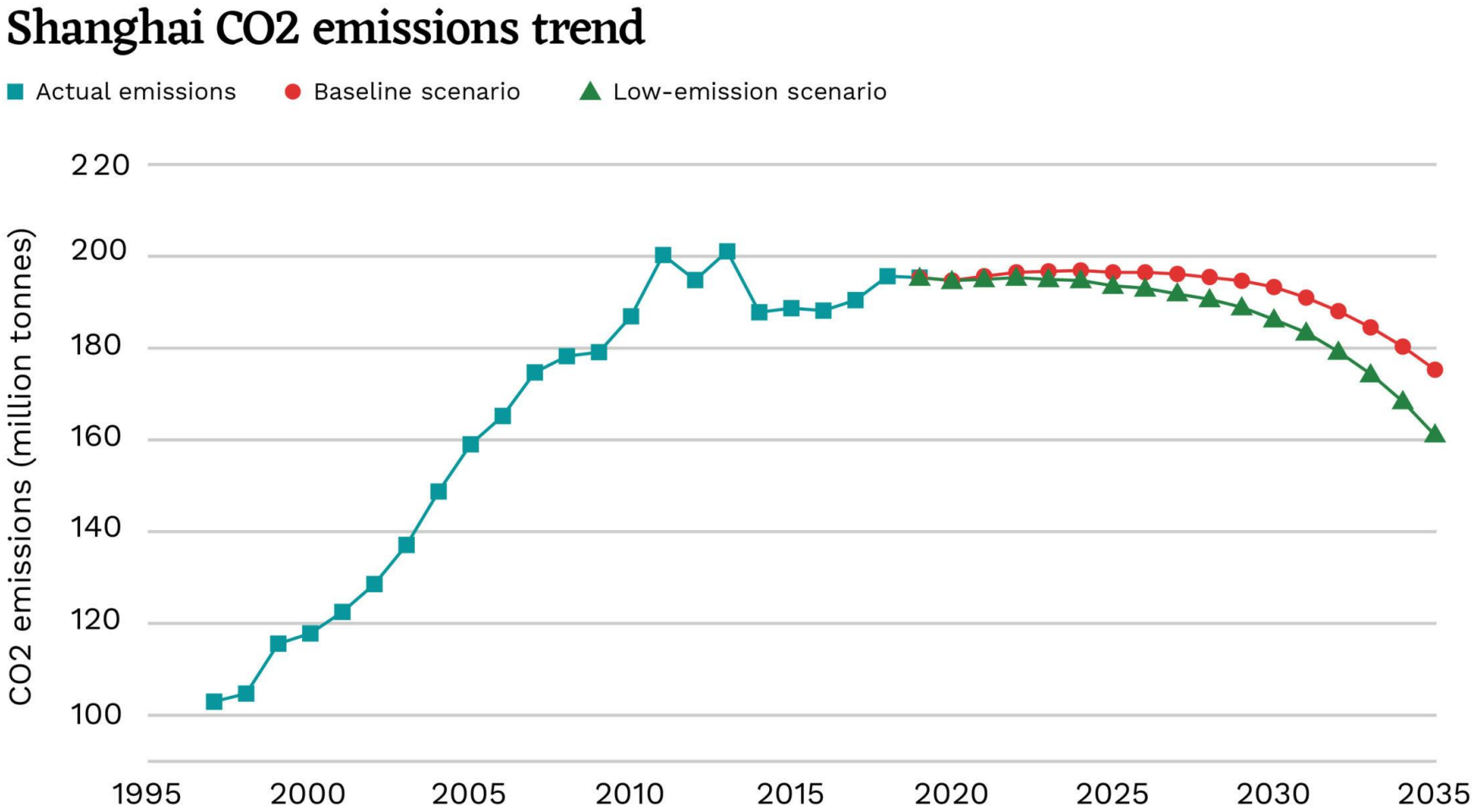


图5 低碳试点常态化温室气体清单编制情况

Example of tracking progress and setting targets



Source: Zhang Zhe et al. "Urban CO2 emissions peaking and low-carbon development: can Shanghai achieve its peaking target?", Environmental Engineering, April 2021.

https://chinadialogue.net/en/cities/shanghai-leads-way-in-chinas-carbon-transition/

中外对话 China Dialogue

02

What Is Being Counted

Defining the city and tracing cross-boundary activities

What is “the city” ?

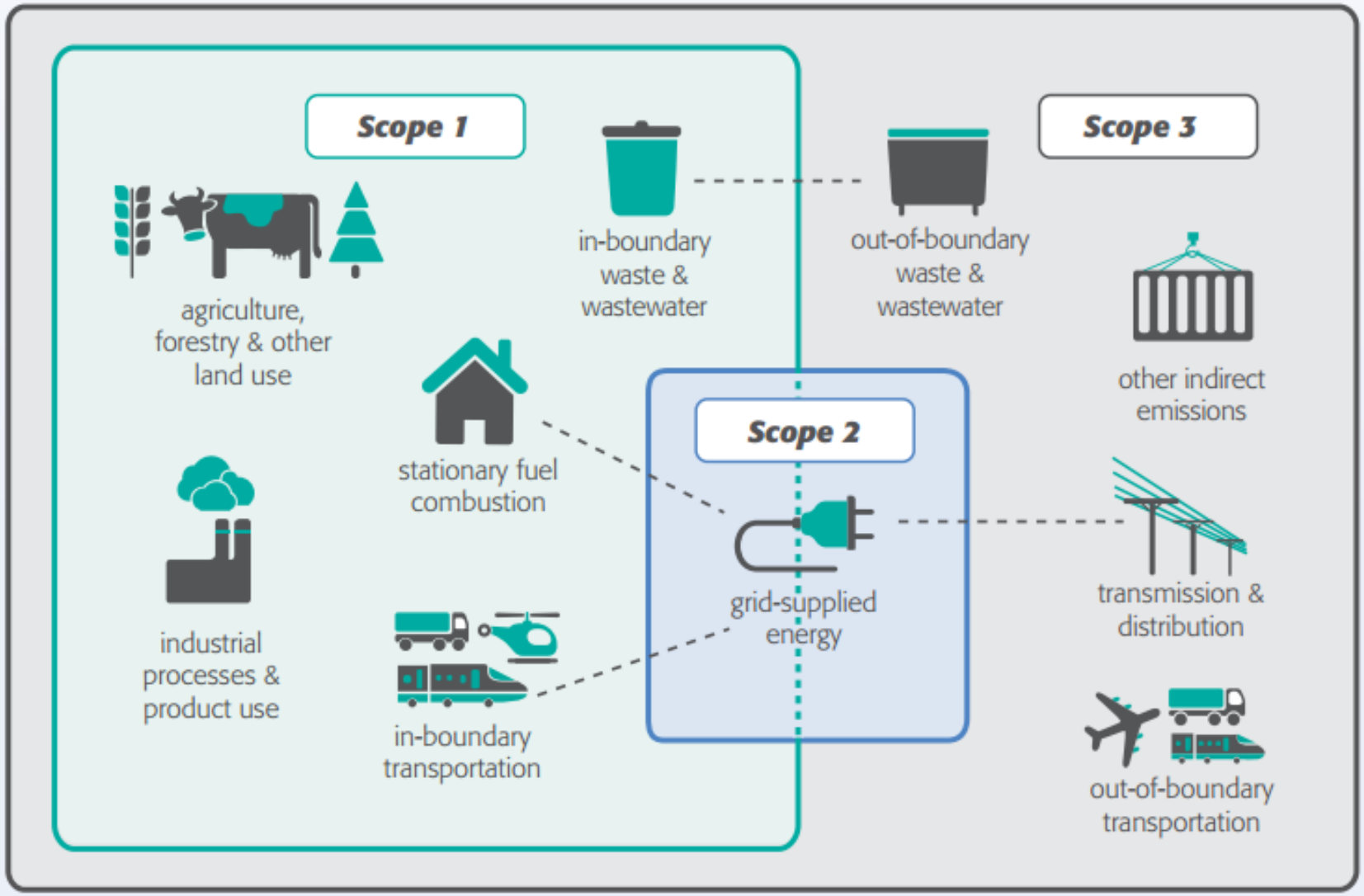
- Administrative city
- Built-up urban area (translate：建成区)
- Metropolitan or functional urban area

Would these three boundaries produce the same emissions result?

City – An open system

- Consider the following activities:
  - Fuel burned in a building inside the city, e.g., natural gas boiler;
  - Electricity generated outside but consumed inside;
  - Waste generated in the city but treated outside;
  - A transport journey crossing the city boundary.

- Some questions to consider:
  - Where do the emissions physically occur?
  - Is the activity connected to the city?
  - Should the city inventory make the emissions visible?
  - What additional information would be needed?


Scope 1
Scope 2
Scope 3
agriculture, forestry & other land use
in-boundary waste & wastewater
out-of-boundary waste & wastewater
other indirect emissions
stationary fuel combustion
industrial processes & product use
grid-supplied energy
transmission & distribution
in-boundary transportation
out-of-boundary transportation

Discussion

Why cities need dedicated accounting guidelines?

(Put it differently, If a city is part of a country, why not simply apply the national inventory framework within the city boundary?.)

City boundaries are challenging.

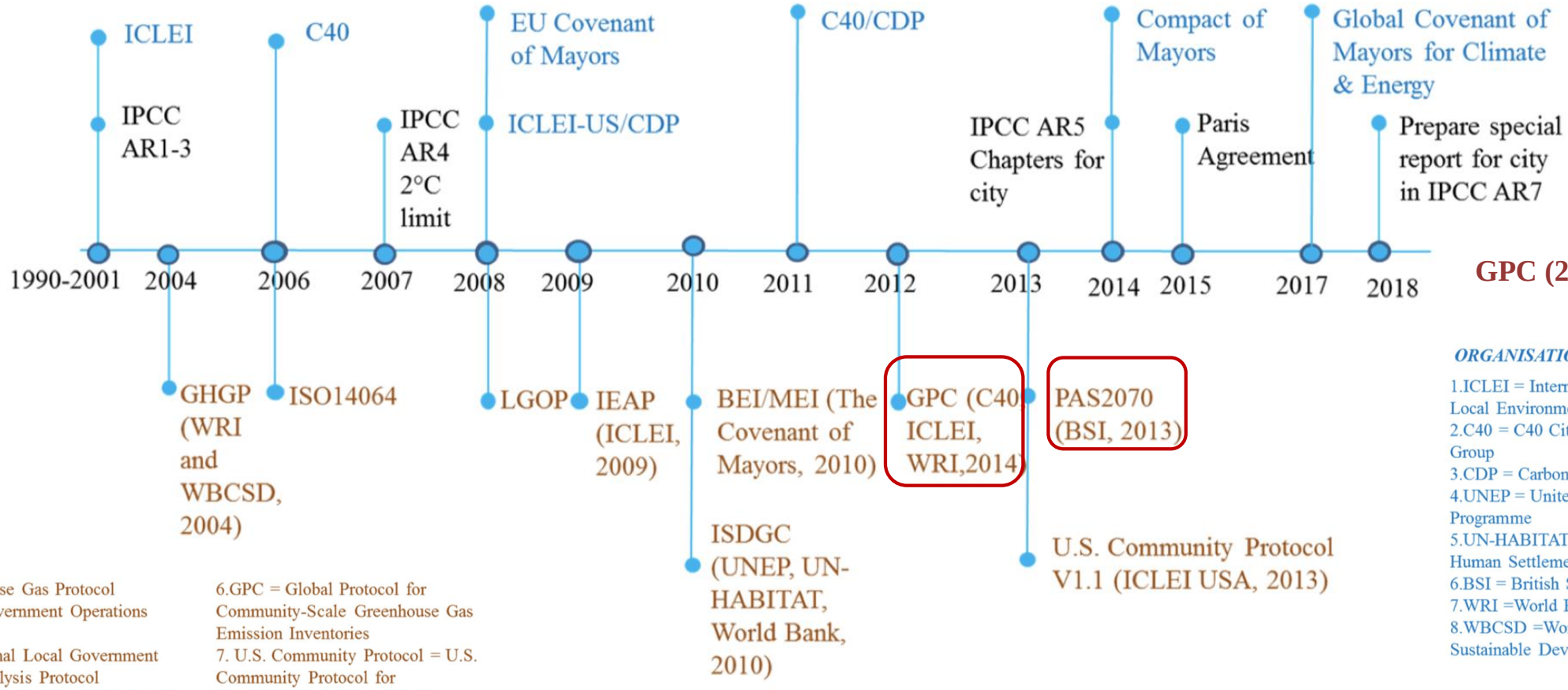
Imported electricity Cross-boundary commuting Imported goods and services

Waste treatment outside the city …

From IPCC Guidelines to City Inventory

What can be reused?	What cities still need to address
Activity data × emission factors	Imported electricity
IPCC sector-specific methods	Cross-boundary transport
Gases and CO_2e	Waste treated outside the city
Data-quality principles	Administrative vs functional boundaries
Territorial emissions	City-specific purposes and actions

City/Community inventories: practitioner view



GPC (2021)

PROTOCOLS

1.GHGP = Greenhouse Gas Protocol
2.LGOP = Local Government Operations Protocol
3. IEAP = International Local Government GHG Emissions Analysis Protocol
4. BEI/MEI = Baseline Emissions Inventory/ Monitoring Emissions Inventory Methodology
5.ISDGC=International Standard for Determining Greenhouse Gas Emissions for Cities
6.GPC = Global Protocol for Community-Scale Greenhouse Gas Emission Inventories
7. U.S. Community Protocol = U.S. Community Protocol for Accounting and Reporting of Greenhouse Gas Emissions
8. PAS 2070= Publicly Available Specification 2070

ORGANISATIONS

1.ICLEI = International Council for Local Environmental Initiatives
2.C40 = C40 Cities Climate Leadership Group
3.CDP = Carbon Disclosure Project
4.UNEP = United Nations Environment Programme
5.UN-HABITAT = United Nations Human Settlements Programme
6.BSI = British Standards Institution
7.WRI =World Resource Institution
8.WBCSD =World Business Council For Sustainable Development

*Geographic PB and CIF methods are included in city protocols 3-8
*CB method is included in city protocols 7,8

From: Chen et al. (2019) EST

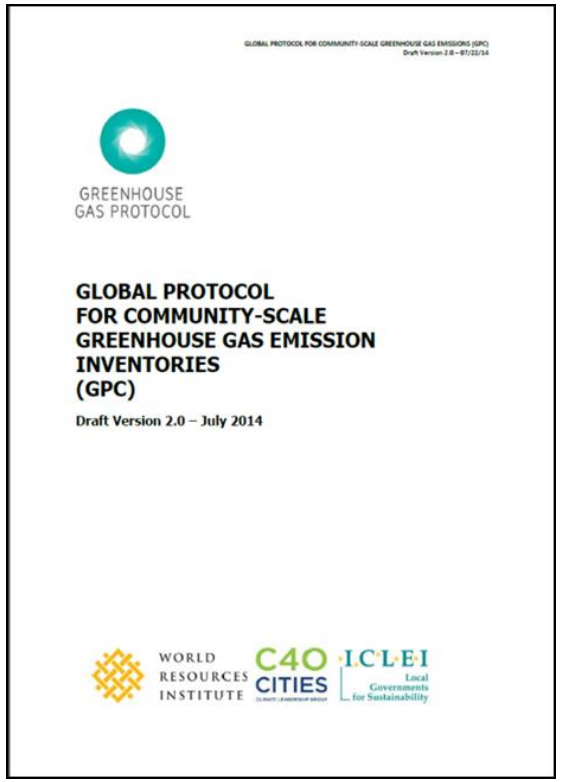
03

How Are City Inventories Built

The GPC framework for sectors, scopes and reporting

City/Community inventories: guidance

- Global Protocol for Community-Scale GHG emissions (GPC)



- PAS 2070 - Specification for the assessment of greenhouse gas emissions of a city (*)

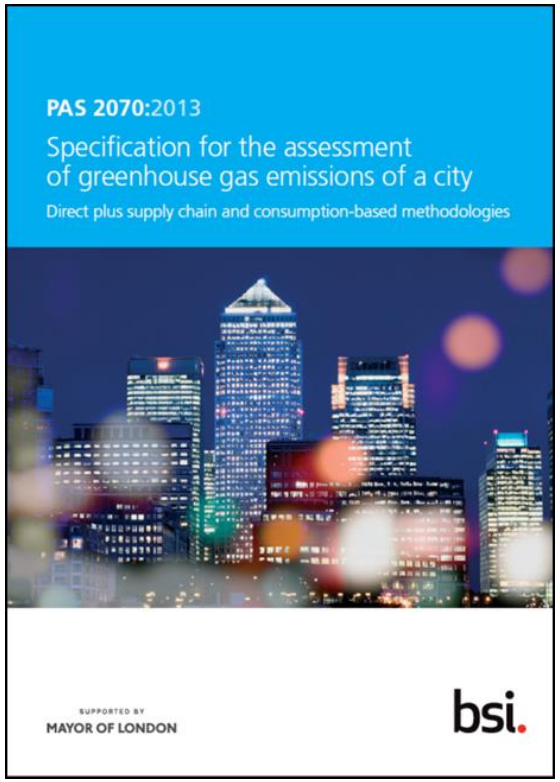
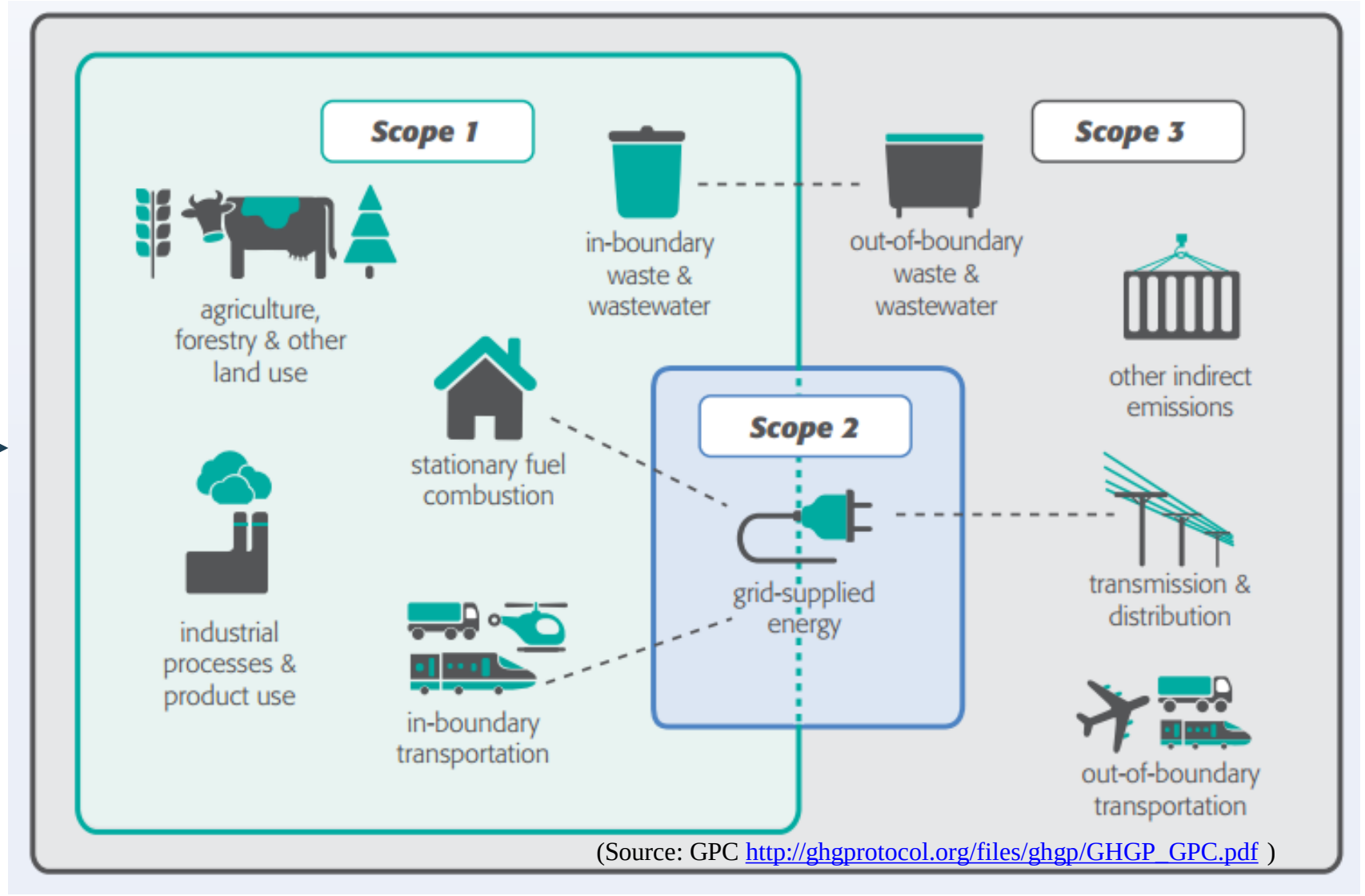


Table 1.4 GPC development process

Date		Milestone
2011	June	Memorandum of Understanding between C40 and ICLEI
2012	March	GPC Draft Pilot (Version 0.9) released for public comment
	May	GPC Draft Pilot (Version 1.0) released
2013		Pilot testing with 35 cities worldwide
2014	July	GPC Draft (Version 2.0) released for public comment
	December	Final GPC published
2021		GPC v1.1 published

City/Community inventories: GPC boundary

- The geographic area
- Time span
- Gases
- Emission sources/sinks



(Source: GPC http://ghgprotocol.org/files/ghgp/GHGP_GPC.pdf)

Scope 1

Scope 2

Scope 3

Table 3.1 Sectors and sub-sectors of city GHG emissions

Sectors and sub-sectors
STATIONARY ENERGY
Residential buildings
Commercial and institutional buildings and facilities
Manufacturing industries and construction
Energy industries
Agriculture, forestry, and fishing activities
Non-specified sources
Fugitive emissions from mining, processing, storage, and transportation of coal
Fugitive emissions from oil and natural gas systems
AGRICULTURE, FORESTRY AND OTHER LAND USE (AFOLU)
Livestock
Land
Aggregate sources and non-CO_2 emission sources on land
INDUSTRIAL PROCESSES AND PRODUCT USE (IPPU)
Industrial processes
Product use
TRANSPORTATION
On-road
Railways
Waterborne navigation
Aviation
Off-road
WASTE
Solid waste disposal
Biological treatment of waste
Incineration and open burning
Wastewater treatment and discharge
OTHER SCOPE 3

Box 4.1 Emission sources and scopes in BASIC and BASIC+

Emission sources and scopes included in **BASIC** totals:

- All scope 1 emissions from *Stationary Energy* sources (excluding energy production supplied to the grid, which shall be reported in the scope 1 total)
- All scope 1 emissions from *Transportation* sources
- All scope 1 emissions from *Waste* sources (excluding emissions from imported waste, which shall be reported in the scope 1 total)
- All scope 2 emissions from *Stationary Energy* sources and transportation
- Scope 3 emissions from treatment of exported waste

BASIC+ totals include all BASIC sources, plus:

- All scope 1 emissions from *IPPU*
- All scope 1 emissions from *AFOLU*
- Scope 3 emissions from *Stationary Energy* sources (only transmission and distribution losses), and from *Transportation*

Source: GPC

Revisit: City – An open system

Activity	Emission location	GPC scope	Sector
Gas burned in an urban building	Inside	Scope 1	Stationary energy
Imported electricity	Outside	Scope 2	Stationary energy
Waste treated outside	Outside	Scope 3	Waste
Cross-boundary metro journey	Inside and outside	Depends on allocation	Transportation

Data and Calculation

Sector	Possible activity data
Buildings	Electricity, gas and heating records
Transport	Fuel sales, traffic counts, vehicle kilometres
Waste	Waste quantity, composition and treatment method
Wastewater	Population, wastewater volume and treatment
Industry	Energy use, output and process data

Top-down data: statistics, utility and departmental data;

Bottom-up data: facility records, surveys and direct measurement;

Proxy data: population, floor area, vehicle ownership and similar indicators.

投票 最多可选1项

设置

A student in City B uses electricity generated by a power plant in City A. If both cities report the emissions, is this double counting?

A Yes—the same emissions must never appear in two city inventories.

B No—the emissions are reported in different scopes, so aggregation is always valid.

C It depends on how the inventories are compared or aggregated.

D No—city inventories do not need to consider double counting.

Choose the approach you currently find most defensible. Be ready to explain why.

提交

City/Community inventories: PAS 2070 (*)

- **City boundary: border of a city or urban area**
  - The city boundary is usually geopolitical and defined by one or more municipal governments.

- **Two different boundary-setting methods:**
  - **Direct Plus Supply Chain** (DPSC)
    - Territorial emissions
    - Main supply-chain emissions (for both production and final consumption)
  - **Consumption-based** (CB)
    - GHG emissions associated with all goods and services consumed by residents of a city

London GHG Emissions in 2010

Table 7 – High level comparison of total GHG by methodology

Methodology	LEGGI [16]	Direct plus supply chain (DPSC)	Consumption-based (CB)
Emission sources	Energy use in buildings, inboundary transport	Energy use in buildings, inboundary and transboundary transport, industrial processes and product use, land use, waste, water, food and construction materials	Supply emissions from consumption of goods and services
Scopes	1, 2	1, 2, 3	
GHG	CO_2	CO_2, CH_4, N_2O, HFCs, PFCs and SF_6	
Total	44.44 $mtCO_2e$	81.06 $mtCO_2e$	114.10 $mtCO_2e$
Per capita	5.51 tCO_2e/capita	10.05 tCO_2e/capita	14.15 tCO_2e/capita

City/Community inventories: methodological discussion*

- There are three main methods for city-level carbon accounting related to socio-economic activities:
  - Territorial-based emissions (TEs) (or pure geographic production based)
  - Consumption-based footprint (CBF)
  - Community Infrastructure-based footprint (CIF)

- The relationship of TEs, CBF, CIF and the Scope 1-3 ?

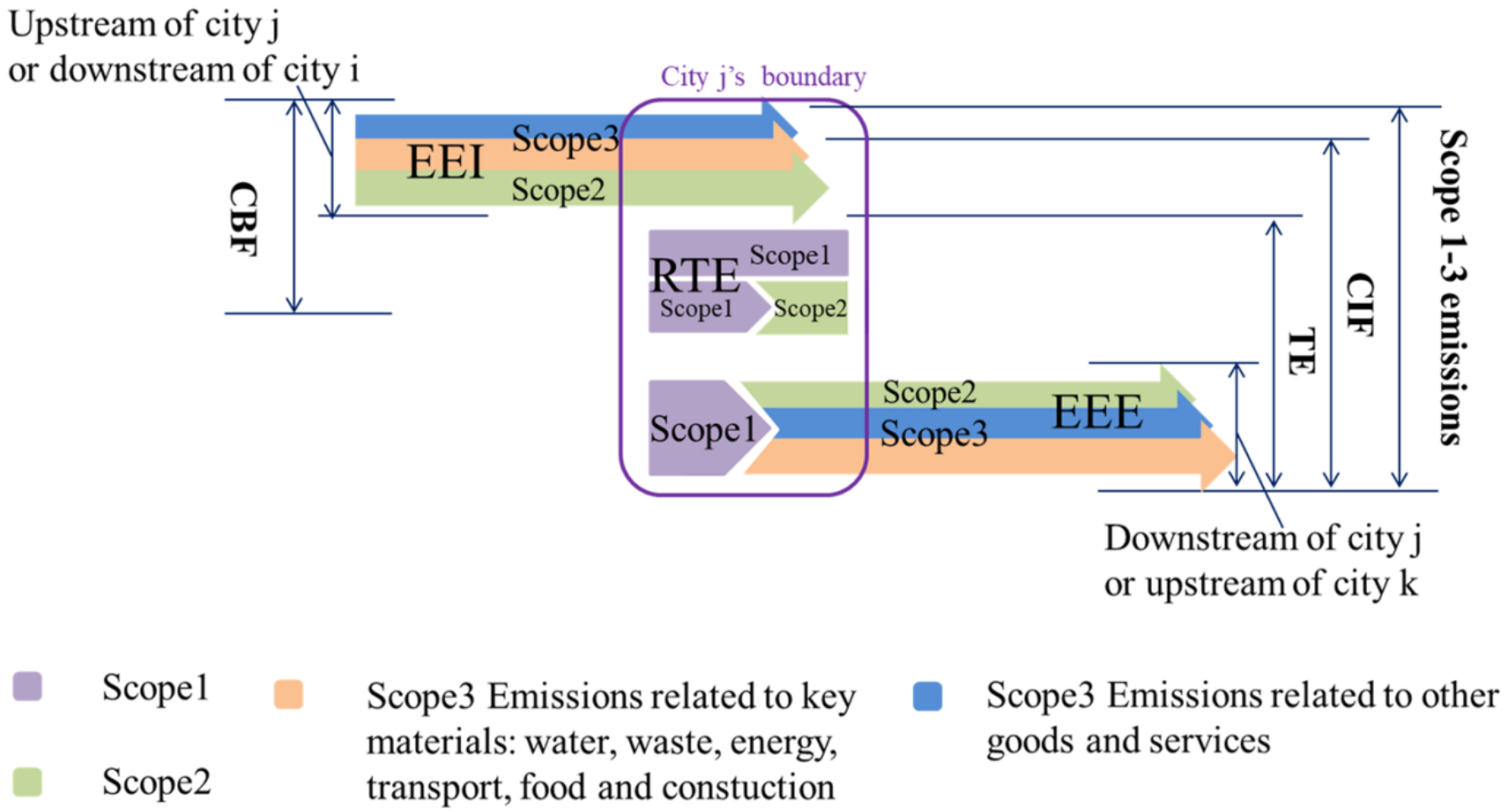


Figure 2. The relationship analysis for TE, CIF, and CBF. CBF = consumption-based carbon footprint (CB method); CIF = community-wide infrastructure footprint (CIF method); TE = territorial emissions (pure-geographic PB method); scope 1–3 emissions = complete scope 1–3 emissions defined in city protocols; EEI = emissions embodied in imports; EEE = emissions embodied in exports; RTE = rest of territorial emissions.

From: Chen et al. (2019) EST.

04

Quiz, Wrap-up and Next Time

Wrap up – understanding a city GHG inventory

Why – why is it produced?

Urban vulnerability and capacity;

Planning, tracking and reporting …

What – what is counted?

A defined geographic boundary

Gases, sectors

Reporting year

Activities and emissions within and across that boundary.

How – how is it quantified?

GPC guidelines

Activity Data * Emission Factors

Statistics/Data

So What – how to interpret it?

Comparability

Model/Method choices

Data differences

A city inventory connects where emissions occur, which urban activities they support, and how the results can inform local action.

NEXT: What changes when the accounting subject is an organization rather than a geographic place?

Next Time – Organizational Level Carbon Accounting

- Essential reading:
  - Corporate Standard | Greenhouse Gas Protocol
    - For Lecture 7: read Chapter 3, pp. 17–18, and Chapter 4, pp. 25–27.
      Focus on organizational boundaries, direct and indirect emissions, and Scope 1, 2 and 3.
  - Lu S., Serafeim G. and Tofffel M.(2022). Driving Decarbonization at BMW. HBS 9-123-008. https://hbsp.harvard.edu/product/ (Available on Canvas. Please do not circulate!)
    - For Lecture 8: read pp. 7–9 and examine Exhibit 11 (p. 25).
      Focus on BMW's emissions profile, data sources and measurement challenges.
- Recommended reading
  - GHG Protocol: Corporate Value Chain (Scope 3) Standard
    - Read the short overview on the webpage and examine the value-chain emissions diagram.
- Further reading:
  - ISO/TS 14064-4:2025 - Greenhouse gases — Part 4: Guidance for the application of ISO 14064-1





See you next time